

Reflection Report

Part2:

The unstable and low acc on part2 was caused by the unstable algo and the badly generated map and the slippery mechanism.

First, let's discuss the unstable acc.

So the first thing we need to do is find a stable algo. to encounter this problem. then we choose the value iteration algo to substitute the previous one.

Second, the low acc was based on two reasons:

One is the slippery mechanism, and the other is the badly generated map .

so if we want to get the higher score is to fix either one of them:

- slippery mechanism:
 - fix the success rate, or just turn off this mechanism
- badly generated map:
 - there are two ways to deal with:
 - one is written the whole map by yourself
 - The second is to find the good map with the seed.

1. Overall, we almost got 90% acc on changing the algo and the different seed.

Part3:

1. In very beginning to solo mode, we let two bot find the kaggle without working together. Once the bot find the kaggle, the searching will be end. The reason is our model let two bot walking in turn, so both of them has to move. Because of that, solo model almost always had a better step than together mode. The reason is solo bot avoid the worse situation.
2. In our result, we find out that the together mode has about two to three steps better than solo mode in many map size. This is a very strange and interesting phenomenon. In our discussion, we think that the reason is the advantage of geometry searching which means two bot searching together and collision cost cancel each other out. Because Bot B's strategy is accessing Bot A, Bot B's accounting more steps with map growing larger. Because of the reason above, we got a small difference between two mode.
3. During the experiment, the together mode cost lots of time to execute. The reason is in together mode, bot A regards bot B as a block and bot A and B keep moving which led to high time complexity because two bot are dynamic block.
4. In our project, our collaborating algorithm has many disadvantage and has a low efficiency compared to solo mode. If we want to improve the result, the collaborating algorithm is the very first to adjustment.